

Output 1 - Morphological Analysis Error in Biomedical Dataset

```
Total unique words: 8473

Word -> Porter Stem
acknowledgment -> acknowledg
becoming -> becom
swedish -> swedish
widening -> widen
nonurban -> nonurban
exacerbation -> exacerb
former -> former
y -> y
prove -> prove
significant -> signific
effort -> effort
utilizing -> util
observations -> observ
nvas -> nva
mohs -> moh
sad -> sad
sensitized -> sensit
hd -> hd
calcitriol -> calcitriol
pirarubicin -> pirarubicin
affiliated -> affili
anorgasmia -> anorgasmia

--- Biomedical Morphology Analysis ---
infection -> infect
infectious -> infecti
infected -> infect
infect -> infect
infections -> infect
infecting -> infect
infective -> infect

--- Words grouped by Stem ---
infect : ['infection', 'infected', 'infect', 'infections', 'infecting', 'infective']
infecti : ['infectious']
```

Output 2 - Error Analysis of Finite-State Morphological Parser in Social Media Data

WORD	OLD FST	MODIFIED FST
happiest	['happi', 'est']	['happy', 'happi', 'est']
unbelievable	['un', 'believable']	['un', 'believ', 'able']
running	['runn', 'ing']	['run', 'runn', 'ing']
reordering	['re', 'ordering']	['re', 'order', 'ing']
smartphones	['smartphone', 's']	['smartphone']
unreadable	['un', 'readable']	['un', 'read', 'able']

Accuracy before correction: 0.0 %
Accuracy after correction: 33.33333333333333 %

Output 3 - Error Analysis of Stemming in News Article Classification

```
Porter Stemmer Results:
organization -> organ
organizer -> organ
organizing -> organ
organized -> organ
organization's -> organization'
```

--- Without Stemming ---

Accuracy: 85.7 %

Confusion Matrix:

```
[[225 15 12 16]
 [ 6 254 2 12]
 [ 18 3 152 32]
 [ 6 4 17 226]]
```

--- Porter Stemming ---

Accuracy: 86.5 %

Confusion Matrix:

```
[[229 15 12 12]
 [ 6 256 2 10]
 [ 20 2 153 30]
 [ 6 4 16 227]]
```

--- Lemmatization ---

Accuracy: 86.2 %

Confusion Matrix:

```
[[228 15 14 11]
 [ 7 255 2 10]
 [ 21 2 153 29]
 [ 8 2 17 226]]
```

Output 4 - Morphological Error Analysis in an Information Retrieval System

```
WORD -> PORTER STEM
-----
i -> i
bought -> bought
two -> two
watches -> watch
and -> and
i -> i
am -> am
watching -> watch
the -> the
product -> product
review -> review
this -> thi
washable -> washabl
cloth -> cloth
is -> is
useful -> use
with -> with
the -> the
washer -> washer
the -> the
clothes -> cloth
were -> were
washed -> wash
yesterday -> yesterday
```

```

MORPHOLOGICAL ANALYSIS
-----
watches -> watch -> Inflectional
watching -> watch -> Inflectional
washable -> washabl -> Derivational
washer -> washer -> Derivational
washed -> wash -> Inflectional

STEM GROUPS
-----
watch : ['watches', 'watching']
washabl : ['washable']
washer : ['washer']
wash : ['washed']

IMPROVED PREPROCESSING
-----
watches -> watche
watching -> watch
washable -> washable
washer -> washer
washed -> wash

```

Output 5 - Identify the Error, Rectify It, and Analyze the Output

```

ORIGINAL WORDS -> STEMMED WORDS
-----

Original : The organization is organizing a new business project.
Stemmed  : the organ is organ a new busi project

Original : The organizer announced the development of new technology.
Stemmed  : the organ announc the develop of new technolog

Original : The government is developing new policies.
Stemmed  : the govern is develop new polici

Original : The companies are connected with international organizations.
Stemmed  : the compani are connect with intern organ

Original : The manager is managing the business operations.
Stemmed  : the manag is manag the busi oper

WORD -> PORTER STEM
-----
a -> a
announced -> announc
are -> are
business -> busi
companies -> compani
connected -> connect
developing -> develop
development -> develop
government -> govern
international -> intern

```

operations -> oper
organization -> organ
organizations -> organ
organizer -> organ
organizing -> organ
policies -> polici
project -> project
technology -> technolog
the -> the
with -> with

20+ STEMMING CASES

organization -> organ
organizer -> organ
organizing -> organ
organized -> organ
political -> polit
politician -> politician
politics -> polit
easily -> easili
studies -> studi
studying -> studi
studied -> studi
relational -> relat
national -> nation
nationality -> nation
connection -> connect
connected -> connect
development -> develop
developing -> develop
developer -> develop

MORPHOLOGICAL ANALYSIS

studies -> studi -> Inflectional
studying -> studi -> Inflectional
studied -> studi -> Inflectional
computers -> comput -> Inflectional
organizing -> organ -> Inflectional
organized -> organ -> Inflectional
developing -> develop -> Inflectional
connected -> connect -> Inflectional
political -> polit -> Derivational
politician -> politician -> Derivational
nationality -> nation -> Derivational
organization -> organ -> Derivational
organizer -> organ -> Derivational
relational -> relat -> Derivational
connection -> connect -> Derivational
development -> develop -> Derivational
developer -> develop -> Derivational
management -> manag -> Derivational
manager -> manag -> Derivational
computerized -> computer -> Derivational
business -> busi -> Derivational

Output 6 - Error Analysis of a Finite-State Morphological Parser

WORD	BASE FORM	TYPE
cars	car	Plural Noun
boxes	box	Plural Noun
cities	city	Plural Noun
children	child	Plural Noun
dogs	dog	Plural Noun
buses	bus	Plural Noun
watches	watch	Plural Noun
men	man	Plural Noun
women	woman	Plural Noun
mice	mouse	Plural Noun
books	book	Plural Noun

WORDNET VERIFICATION

cars	-> car
boxes	-> box
cities	-> city
children	-> child
dogs	-> dog
buses	-> bus
watches	-> watch
men	-> man
women	-> woman
mice	-> mouse
books	-> book

Output 7 - Error Analysis of Morphological Feature Extraction

ORIGINAL WORDS -> STEMMED WORDS

Original : The organization is organizing a new business project.
Stemmed : the organ is organ a new busi project

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Stemmed : the organ announc the develop of new technolog

Original : The government is developing new policies.
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Original : The companies are connected with international organizations.
Stemmed : the compani are connect with intern organ

Original : The manager is managing the business operations.
Stemmed : the manag is manag the busi oper

WORD -> PORTER STEM

a -> a
announced -> announc
are -> are
business -> busi
companies -> compani
connected -> connect
developing -> develop
development -> develop
government -> govern
international -> intern
is -> is
manager -> manag
managing -> manag
new -> new
of -> of
operations -> oper
organization -> organ
organizations -> organ
organizer -> organ

20+ STEMMING CASES

organization -> organ
organizer -> organ
organizing -> organ
organized -> organ
political -> polit
politician -> politician
politics -> polit
easily -> easili
studies -> studi
studying -> studi
studied -> studi
relational -> relat
national -> nation
nationality -> nation
connection -> connect
connected -> connect
development -> develop
developing -> develop
developer -> develop
management -> manag
manager -> manag
computers -> comput
computerized -> computer
business -> busi

MORPHOLOGICAL ANALYSIS

studies	-> studi	-> Inflectional
studying	-> studi	-> Inflectional
studied	-> studi	-> Inflectional
computers	-> comput	-> Inflectional
organizing	-> organ	-> Inflectional
organized	-> organ	-> Inflectional
developing	-> develop	-> Inflectional
connected	-> connect	-> Inflectional
political	-> polit	-> Derivational
politician	-> politician	-> Derivational

nationality	-> nation	-> Derivational
organization	-> organ	-> Derivational
organizer	-> organ	-> Derivational
relational	-> relat	-> Derivational
connection	-> connect	-> Derivational
development	-> develop	-> Derivational
developer	-> develop	-> Derivational
management	-> manag	-> Derivational
manager	-> manag	-> Derivational
computerized	-> computer	-> Derivational
business	-> busi	-> Derivational